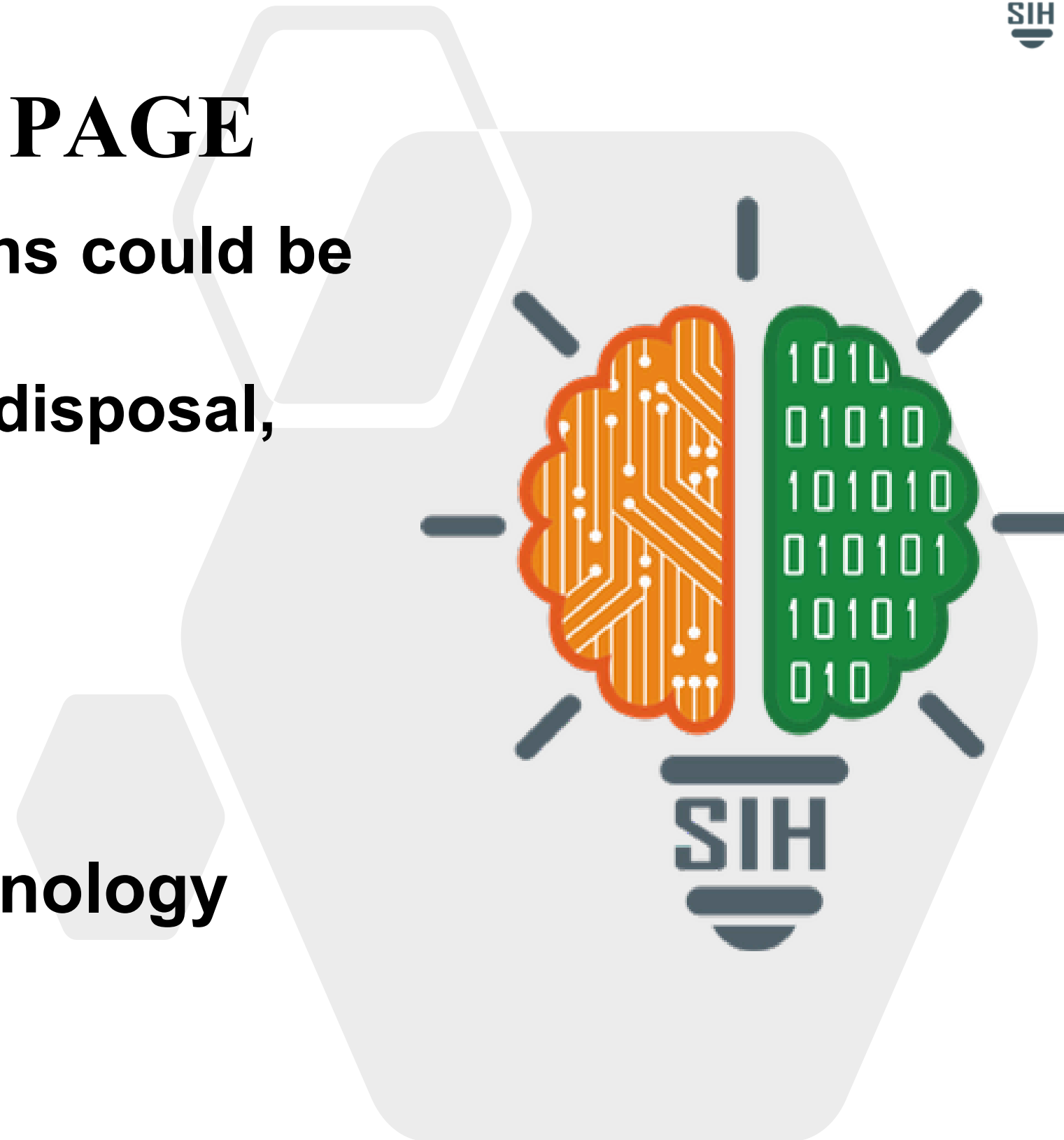


TITLE PAGE

- **Problem Statement Title** - Solutions could be in the form of waste segregation, disposal, and improve sanitization system.
- **Problem ID** - 26195
- **Theme** - Clean and Green Technology
- **PS Category** - Software
- **Team name** - Deltz Shot





Greenox “The Digital System for Transparent Sanitation Issue Resolution”

TEAM:- DELTZ SHOT

REPORT → VALIDATE → SMART ASSIGN → CLEAN → VERIFY → PREVENT

• A verified sanitation operations platform that converts citizen reports into tracked, verified and accountable cleanup.

Problem Addressed

- Reports can be duplicate , incomplete or wrongly routed.
- Duplicate complaints increase workload.
- Difficult to track genuine completion and recurring hotspots.

Our Solution

- Citizen submits photo + location + category.
- System validates and identifies possible duplicates.
- Officer assigns work according to team availability and workload.
- Worker submits genuine completion evidence.
- Citizen/supervisor can verify, reject or reopen.
- Repeated locations become hotspot history.

Innovation

- Verified Closure Loop — Report → Clean → Verify .
- Smart Duplicate Detection — Reduce repeated complaints & visits.
- Capacity-Aware Assignment - Right task → Right team → Right time.
- Tamper-Aware Verification — Evidence-backed, auditable closure
- Hotspot Intelligence — Detect & prevent recurring sanitation issues
- AI + Human Review — AI-assisted decisions with human oversight

“From Every Report to a Verified, Lasting Resolution”



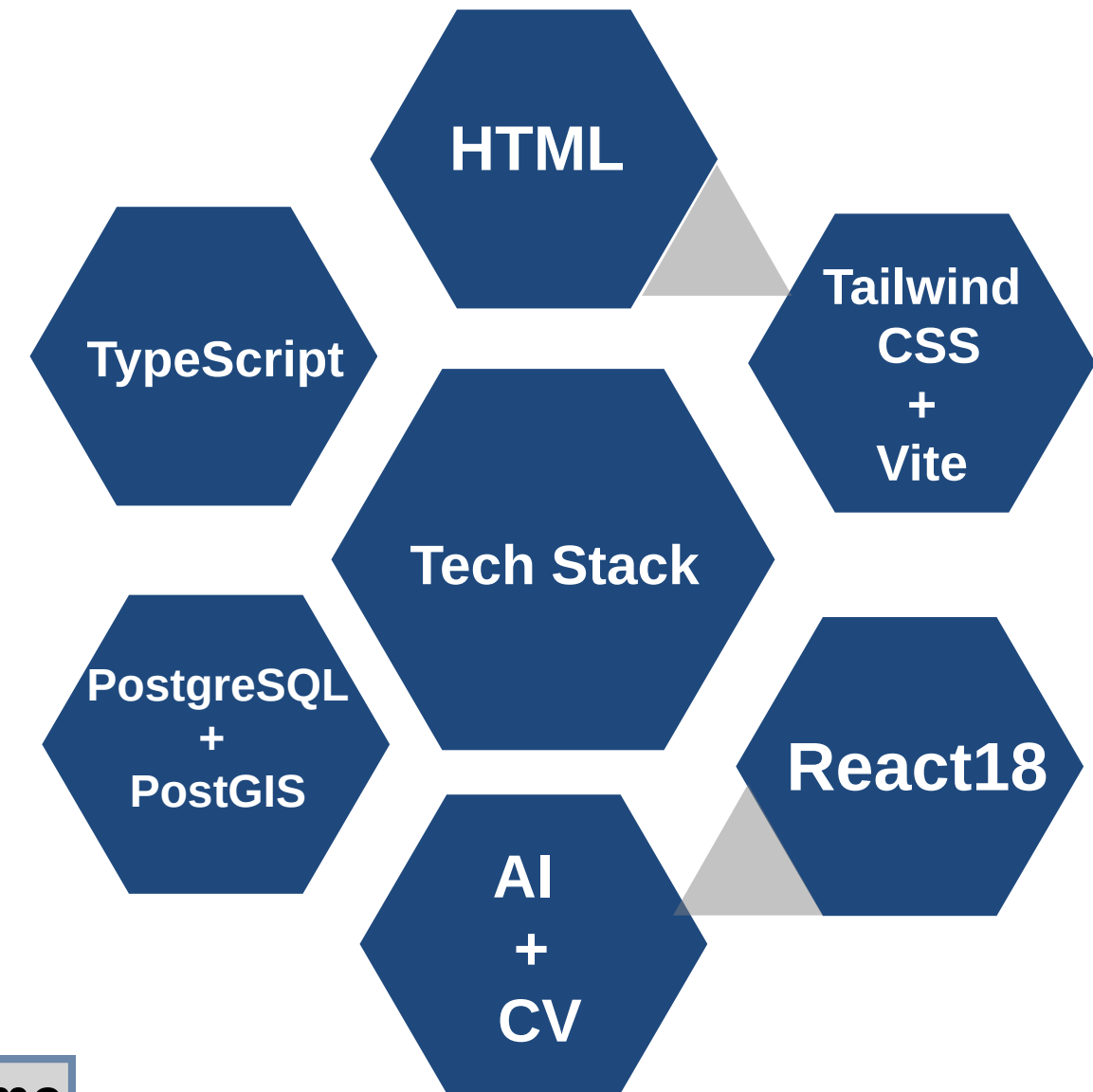
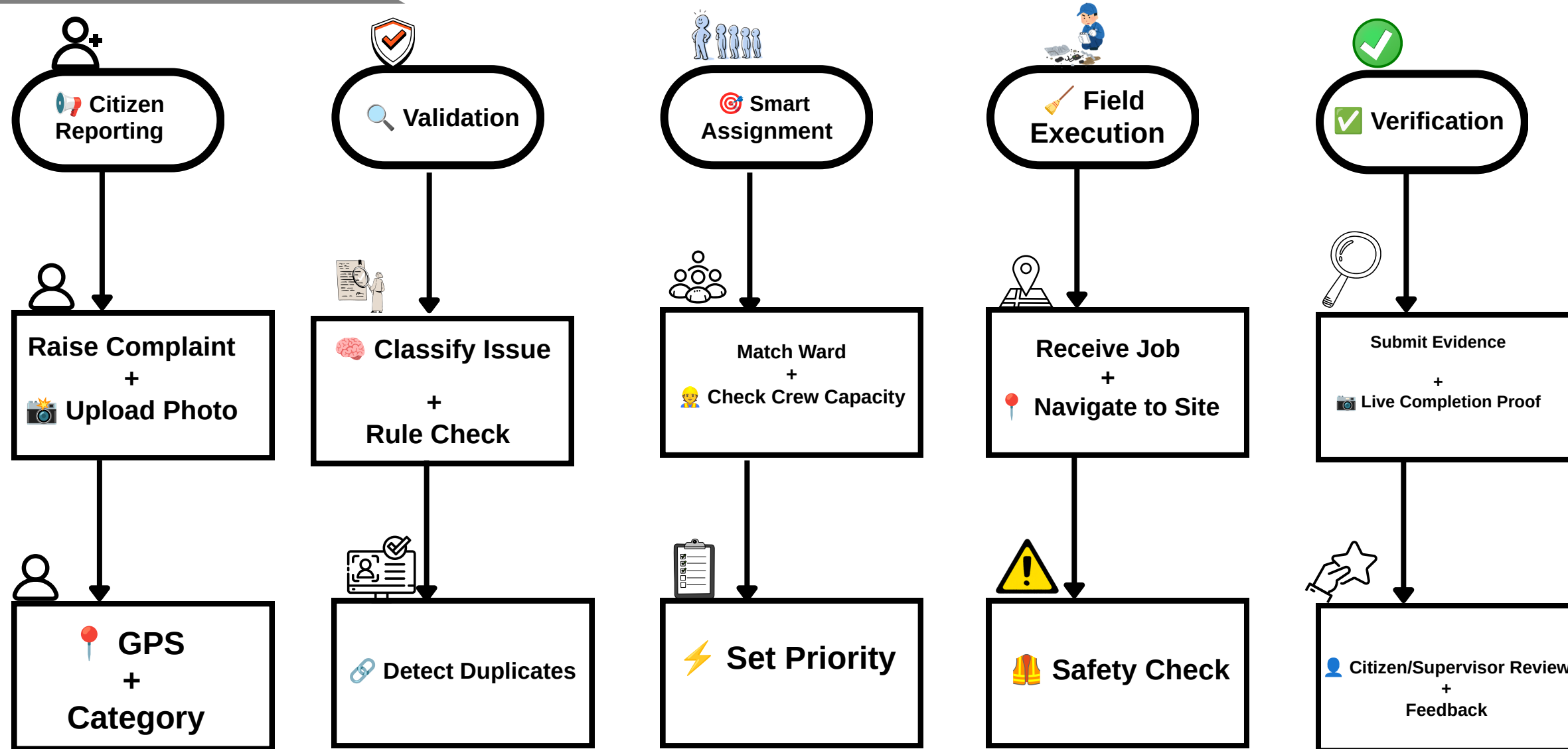
TEAM:- DELTZ SHOT

TECHNICAL APPROACH

Greenox User Flow



SMART INDIA
HACKATHON
2026



Data Captured

- Upload Photo
- GPS

Intelligent Tracking

- Deduplication.
- GIS
- Priority

Verified Outcome

- Evidence
- Feedback



TEAM:- DELTZ SHOT

FEASIBILITY AND VIABILITY



FEASIBILITY (Can we build it?)

- Software-first MVP; no new hardware is mandatory.
- Smartphone camera and GPS provide primary field inputs.
- Pre-trained computer-vision models can bootstrap image analysis.
- Role-based workflow maps directly to citizen, admin and worker modules.
- Cloud APIs support centralized storage, notifications and analytics.
- Pilot can start with one ward and expand after KPI validation

CHALLENGES & RISKS (What can go wrong?)

- Low light, blur, rain and occlusion can reduce model confidence.
- Garbage appearance varies across localities and seasons.
- Poor connectivity may interrupt field updates.
- False positives/negatives can affect priority decisions.
- Citizen images may contain incidental personal information.
- Legacy municipal systems may require integration agreements.

MITIGATION (How do we control risk?)

- Show confidence and keep human override for uncertain cases.
- Build a locally representative, consented training dataset.
- Use offline draft/queue and sync when connectivity returns.
- Keep AI advisory for high-impact decisions.
- Apply data minimisation, access control and retention rules.
- Expose interoperable case IDs and status APIs.

VIABILITY (How will the project sustain itself?)

- 🏛️ Primary Customer: ULBs / Municipal Bodies.
- 💼 Revenue: Paid pilots + annual platform/support contracts.
- 🏢 Institutional Market: Campuses, RWAs, hospitals & industries.
- 📈 Scalability: 2-ward pilot → City → Multi-city.
- 🆓 Citizen Access: Reporting remains free.



TEAM:- DELTZ SHOT



BENEFITS

As a Citizen

- Transparent complaint tracking
- Faster & clearer resolution updates
- Easy reporting with photo + location
- Feedback & reopen facility

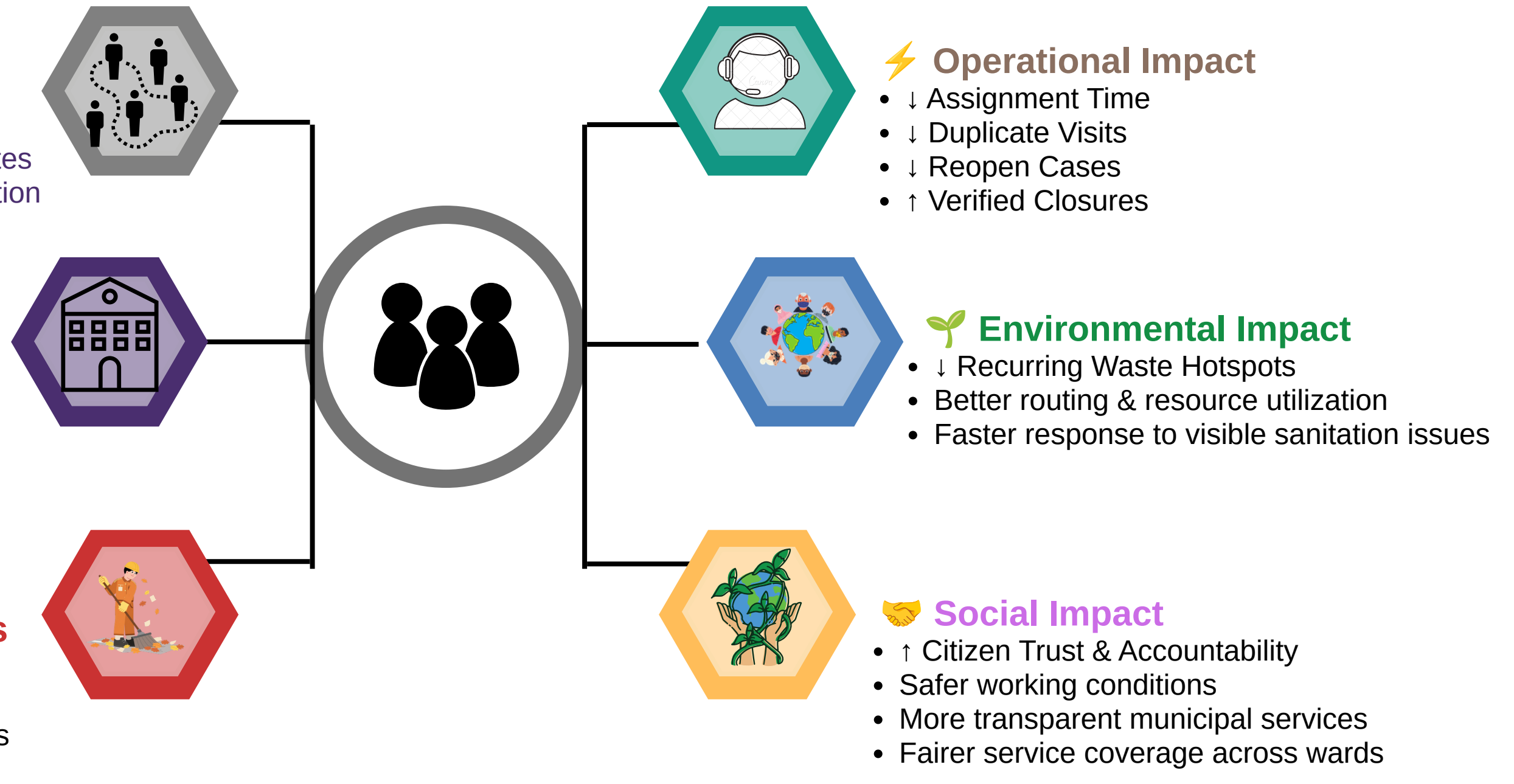
As a Municipal Authorities

- Reduced duplicate field visits
- Smarter task assignment
- Evidence-backed closure
- Better operational visibility

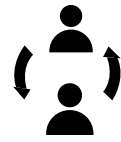
As a Sanitation Workers

- Clear task & location information
- Safety-aware work instructions
- Offline support for field operations
- Easy escalation of unsafe tasks

IMPACT AND BENEFITS



SMART INDIA
HACKATHON
2026

 **IMPACT**

RESEARCH AND REFERENCES

TOPIC	REFERENCE	DRAWBACKS	LINK
Sanitation Grievance	Swachhta-MOHUA,2024	Dependent on municipal response ; limited field-level verification	https://sbmurban.org/
AI Waste Detection	IEEE/ScienceDirect,2023	Sensitive to occlusion , lightning , clutter and unseen waste types	https://www.sciencedirect.com/
Geo-Location & GIS	OpenStreet/Leaflet,2024	Accuracy depends on map data ; geocoding results may be approximate.	https://www.openstreetmap.org/

RESEARCH PAPER STUDIED			
Paper	What is studied	Main lesson for GREENOX	Important limitation
Thakur, Parida, and Raj, 2022/2024 issue	Sustainable municipal waste factors and the Rourkela journey	Municipal waste management is not only about collecting waste — it needs an efficient, organized system to plan, monitor, and improve the entire operation.	One Indian city and model-based factor analysis
Mersico , Aureli and Foschi , 2026	A digital-platform case in Italian municipal waste management	A platform can act as a broker among citizens, city, and waste companies and reduce fragmentation	Single Italian case; context limits generalization
Singh, Singh, and Singh, 2016	A Web-GIS complaint and monitoring framework for solid waste	Open web maps and PostGIS can support citizen reporting and officer monitoring	Older framework; not proof of city-wide impact